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6G Wireless Communication Systems: Emerging Technologies, Architectures, Challenges, and Opportunities

5.1 Introduction

The use of 5G in mobile devices worldwide is growing, and almost everyone can use it. But just as its possibilities are revealed, so too are limitations. Because of this, the idea that we can go further has arisen, leading us to conceptualize and pursue 6G. Emerging 6G technologies would help us further our horizons, as we have never seen before. Artificial intelligence (AI) and the Internet of Everything (IoE) will be deeply affected by this development. Considering how 5G has created smart homes, smart cars, and everything in between, 6G could further create much more. New architectures have been designed for 6G, and there are new challenges and opportunities within this technology. Research on 6G has only been recent, and there is a world of opportunities we could create as the years go by.

Cognitive service architecture is a new architecture designed specifically for the core network, meaning it can potentially enhance the core network to give the highest quality of service (QoS) and different scenarios. Cognitive service architecture would greatly enhance the performance of 6G. There will be frequencies up to terahertz, unlimited bandwidth, and microsecond latency. This new architecture would support all the needs and expectations that 5G was not able to satisfy.

The 6G network should eventually replace the 5G network in the coming years. While 6G makes sense as a successor to 5G, it could never be referred to as such. If it were not for anything like 5G Enhanced or 5G Advanced, we might just say “we're connected” instead of all the numbers and names. Every decade or so, a new mobile network standard becomes the center of attention. This indicates that 6G networks could start rolling out by about 2030, or at the very least when many service providers will be conducting 6G tests, and phone manufacturers will be teasing 6G-capable devices. On a 6G network, anything you do now that requires a network connection would be considerably better. Every advance that 5G delivers will be mirrored in a 6G network as an even superior, advanced version. Statistics demonstrate the importance of strengthening communication networks. We're on our way to a society dominated by fully autonomous remote monitoring systems.

Autonomous systems have gained traction in various fields – notably industry, agriculture, transportation, marine, and aerospace. To create a smart life and automated processes, tons of sensors are incorporated into cities, automobiles, residences, enterprises, nutrition, sports, and other settings. As a result, these applications will want a high-data-rate connection with a high level of reliability. 5G communication protocols have already been installed in several regions of the world and were expected to be completely operational globally by 2022.

5.2 Important Aspects of Sixth-Generation Communication Technology

The transition from analog to digital communication of 2G has paved the road for flexibility by means of reuse frequency options, multiple modulation schemes, adaptive equalization, and dynamic frequency allocation. The third-generation (3G) and fourth-generation (4G) cellular systems incorporated bundles of services such as voice, data, and video streaming. Access technologies such as code division multiplexing (CDMA) and orthogonal frequency division multiplexing (OFDM) have enhanced multiplexing methods, flexible and adaptive rate, and interference management via the utilization of frequency farming and multiple streaming [1].

5.2.1 A Much Higher Data Rate

Technology reports show that 5G is set to deliver 20 Gbps peak data rates and 100 Mbps user data rates. By contrast, the peak data rates for 6G are estimated to be approximately 1000 Gbps. Likewise, the user data rate experience is set to increase to approximately 1 Gbps. Hence, it is estimated that the spectral efficiency of the 6G network will be twice more than that of 6G. This advanced spectral improvement is set to change the face of multimedia services improvement both instantly and simultaneously. Therefore, the 6G network revamp is characterized by a much higher data rate.

5.2.2 A Much Lower Latency

The 5G infrastructure is set to reduce the existing latency by approximately one millisecond. The role of this ultra-low latency is to enhance the performance of various real-time applications. On the other hand, the 6G communication technology is projected to reduce the latency by less than 0.1 ms. Because of the huge decline in latency, real-time applications will realize improved functionality and performance. Besides, lower latency is known to improve emergency response calls, industrial automation, and remote surgeries. In retrospect, 6G networks are projected to make networks 100 times better than 5G networks.

5.2.3 Network Reliability and Accuracy

Without a doubt, the 6G network infrastructure will be more reliable than the 5G network. For instance, 5G is built to support the highest speeds in mobile devices by approximately 500 km/h, whereas 6G is set to serve over 10 devices per square kilometer. This means that many devices are able to interact with each other in real-time effectively. Furthermore, 6G is set to optimize M2M interaction by improving network reliability.

5.2.4 Energy Efficiency

6G is all about promoting real-time interactions seamlessly without any technical interruptions or disruptions. It offers end-users a great experience that is characterized by high-end services and powerful batteries for devices. It means that one of the goals of 6G is to improve the user experience while promoting environmental sustainability because its design uses less energy. This increases its optimization and energy efficiency, making it superior to 5G networks.

5.2.5 Focus on Machines as Primary Users

Both 5G and 6G aim at improving M2M communication, but they will do so at different levels and capacities. For instance, 5G is more focused on mobile and human experience, whereas 6G focuses on machine communication as its primary use. 6G network functionality is set to boost M2M interaction across different devices and machines such as robots, drones, home appliances, and smart sensors. Besides, these wireless networks enable the effective transition into next-generation devices that use technologies such as augmented reality (AR) glasses and holograms.

5.2.6 AI Wireless Communication Tools

Technologists believe that 6G will transform wireless communications, especially those using artificial intelligence. Unlike the 5G networks, the designs for 6G networks aim to leverage the most practical applications of artificial intelligence. Artificial intelligence is a technology that is beneficial in reducing infrastructural costs and improving operational efficiency. The real-time data collected in the connected devices and various communication devices are set to improve services.

5.2.7 Personalized Network Experience

OpenRAN characterizes 5G as an evolving technology, whereas 6G is set to utilize OpenRAN as a mature technology. According to technology engineers, an AI-driven RAN allows personalized network experiences for users dependent on real-time data from various sources.

Based on these characteristics of 6G networks, it is evident that reliable data connectivity is vital to a ubiquitous digital world. The development of networks from 4G to 6G over the years shows the vitality of communication and the roles technologies play in the social and physical world. In addition, technology research predicts that the current deployment of the 5G network will be unable to fulfill the current and future technological demands. Thus, telecommunication engineers are working toward developing and rolling out 6G networks because they are superior in their architecture, scalability, and reliability. Thus, understanding the evolution of 5G to 6G by assessing its technologies, architecture, and uses is critical for utilizing various communication paradigms [2].

Mobile technology generations are often designed to support an end to end-users and various network operators. Figure 5.1 shows how network generations have evolved over time due to physical and human demands. The current technological demands are more data-centric, dependent, and automated. Thus, it requires constant improvement from 4G and 5G technologies to future networks such as 6G. Autonomous systems characterize the world as many industries embrace sensors to operate their activities. For instance, machines embedded with many sensors characterize the world's roads, airspace, and waters and systems operated using artificial intelligence.

Even though the talks surrounding 6G are on the rise, credit should be given to its predecessor, 5G networks, because they have paved the way for further innovation. For instance, there are improvements in frequency bands, latency, spectrum usage, and management brought by the need to develop 6G networks.

Thus, it is rational to state that the development and rolling out of 6G networks are inevitable because they can optimize artificial intelligence and the internet of things systems. The hype to have an intelligent digital world puts pressure on engineers to focus on 6G network architectures. Table 5.1 shows the parameters and features of the latest wireless generations.

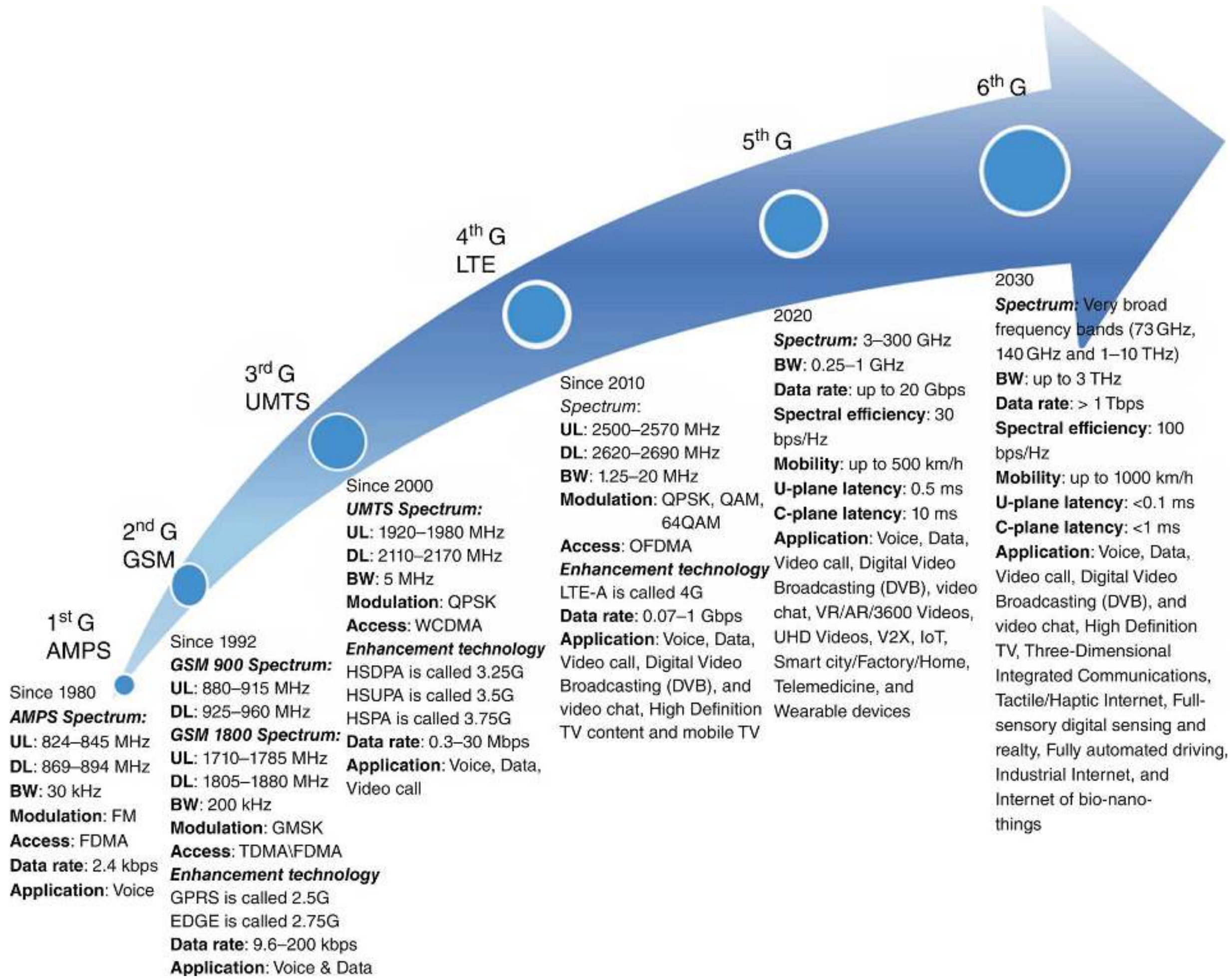

Figure 5.1 Network evolution graph over the years [3].

Table 5.1 5G to 6G parameters and features [4].

Issue	**4G**	**5G**	**6G**
Per device peak data rate	1 Gbps	10 Gbps	1 Tbps
End-to-end (E2E) latency	100 ms	10 ms	1 ms
Maximum spectral efficiency	15 bps/Hz	30 bps/Hz	100 bps/Hz
Mobility support	Up to 350 km/h	Up to 500 km/h	Up to 1000 km/h
Satellite integration	No	No	Fully
AI	No	Partial	Fully
Autonomous vehicle	No	Partial	Fully
XR	No	Partial	Fully
Haptic communication	No	Partial	Fully
THz communication	No	Very limited	Widely
Service level	Video	VR, AR	Tactile
Architecture	MIMO	Massive MIMO	Intelligent surface
Maximum frequency	6 GHz	90 GHz	10 THz

5.3 Enabling Technologies Behind the Drive for 6G

Originally, 6G networks were primarily built on the 5G framework, acquiring the gains acquired in 5G, given the past progress of mobile networks. Various technological (Figure 5.2) advances will be introduced, while others from 5G will be upgraded to 6G. As a result, a variety of technologies would power the 6G infrastructure.

5.3.1 Artificial Intelligence

The basic property of 6G autonomous networks is intelligence. As a result, AI is the most important and recently launched technology for 6G communications networks. AI was not used to develop 4G network technologies. Partially or very restricted AI would be supported by the next 5G communications networks. Nevertheless, AI will significantly assist 6G in automation [6]. The entire capability of radio transmissions will be realized with AI-powered 6G, allowing the transition from cognitive to smart radio. Increasingly smart systems for real-time connectivity in 6G are being created thanks to advances in machine learning (ML). The use of artificial intelligence in communication will make the transmission of real-time data easier and more efficient.

AI can determine how a difficult target job is accomplished using a variety of statistics. AI boosts productivity and minimizes the time it takes for communication stages to be processed.

AI would be used to quickly complete time-consuming processes like transfer and connectivity selection.M2M, machine-to-human, and human-to-machine (H2M) communications will all benefit from AI. It also encourages BCI interaction. Nanostructures, smart structures, intuitive networks, telecommunication technologies, adaptive cognitive radio, conscious wireless systems, and deep learning will all be used to enable ML communication systems. Integrating photonics and AI will accelerate the advancement of AI within 6G, allowing for the development of digital logic cognitive radio networks. AI-based encoder-decoders, deep neural networks for channel state prediction, and automatic modulation categorization are used in the physical layer. Deep learning-based allocation of resources and smart traffic modeling have been intensively investigated to meet the 6G needs for the data link layer and the transport layer [6].

The combination of big data analytics to discover the optimum route to convey data to end-users by giving data modeling would drastically minimize latency. Table 5.2 shows 6G emerging technologies for different services.

Enabling Technology	Potential	Challenges	Use cases
New Spectrum			
Terahertz	High bandwidth, small antenna size, focused beams	Circuit design, high propagation loss	Pervasive connectivity, industry 4.0, holographic telepresence
VLC	Low-cost hardware, low interference, unlicensed spectrum	Limited coverage, need for RF uplink	Pervasive connectivity, eHealth
Novel PHY techniques			
Full duplex	Continuous TX/RX and relaying	Management of interference, scheduling	Pervasive connectivity, industry 4.0
Out-of-band channel estimation	Flexible multi-spectrum communications	Need for reliable frequency mapping	Pervasive connectivity, holographic telepresence
Sensing and localization	Novel services and context-based control	Efficient multiplexing of communication and localization	eHealth, unmanned mobility, industry 4.0
Innovative Network Architectures			
Multi-connectivity and cell-less architecture	Seamless mobility and integration of different kinds of links	Scheduling, need for new network design	Pervasive connectivity, unmanned mobility, holographic telepresence, eHealth
3D network architecture	Ubiquitous 3D coverage, seamless service	Modeling, topology optimization and energy efficiency	Pervasive connectivity, eHealth, unmanned mobility
Disaggregation and virtualization	Lower costs for operators for massively dense deployments	High performance for PHY and MAC processing	Pervasive connectivity, holographic telepresence, industry 4.0, unmanned mobility
Advanced access-backhaul integration	Flexible deployment options, outdoor-to-indoor relaying	Scalability, scheduling, and interference	Pervasive connectivity, eHealth
Energy-harvesting and low-power operations	Energy-efficient network operations, resiliency	Need to integrate energy source characteristics in protocols	Pervasive connectivity, eHealth
Intelligence in the network			
Learning for value of information assessment	Intelligent and autonomous selection of the information to transmit	Complexity, unsupervised learning	Pervasive connectivity, eHealth, holographic telepresence, industry 4.0, unmanned mobility
Knowledge sharing	Speed up learning in new scenarios	Need to design novel sharing mechanisms	Pervasive connectivity, unmanned mobility
User-centric network architecture	Distributed intelligence to the endpoints of the network	Real-time and energy-efficient processing	Pervasive connectivity, eHealth industry 4.0
Not considered in 5G		With new features/capabilities in 6G	

Figure 5.2 Comparison of 6G enabling technologies and relevant use cases [5].

Table 5.2 Characterization of emerging technologies under different 6G services [4].

Technology	uMUB	uHSLLC	mMTC	uHDD
Artificial intelligence	√	√	√	√
Terahertz communications	√	√		
OWC	√	√	√	√
FSO fronthaul/backhaul	√	√		
Massive MIMO		√	√	√
Blockchain		√		
3D networking	√	√		√
Quantum communications		√	√	
Unmanned aerial vehicles	√	√	√	√
Cell-free communications	√	√	√	√
Integration of wireless information and energy transfer	√			√
Integration of sensing and communication	√			√
Integration of access-backhaul networks	√	√		
Dynamic network slicing	√	√		
Holographic beamforming		√		
Big Data analytics		√	√	√
Backscatter communication			√	
Intelligent reflective surface		√	√	√
Proactive caching		√	√	√
Mobile edge computing	√			

5.3.2 Terahertz Communications

The bandwidth can be increased to improve spectral efficiency. Widening bandwidths and employing advanced enormous multiple-input, multiple-output (MIMO) technology are two ways to do this.

5G uses mmWave frequencies to boost data throughput and open up new possibilities. 6G, on the other hand, wants to stretch the frequency band's bounds to THz to fulfill an even larger demand. The RF band is nearly full, but it is no longer able to ensure the high demands of 6G. For 6G communication, the THz band will be critical. The THz band is envisioned as the next step forward in high-data-rate transmission. Perspective 6th generation (6G) wireless networks will be made up of a slew of small radio cells coupled with high-speed communications cables. Wireless transmission at the Terahertz band is a particularly appealing and adaptable approach in this regard. The THz spectrum has been the electromagnetic family's real contender. Not only are they less well-known than radio or microwaves, but there are fewer technologies capable of generating and modulating these waves.

The terahertz realm is a technological dilemma, apart from submillimeter instruments in astronomy, and it is often alluded to as the “terahertz gap.” Terahertz technologies are less well established and hence more expensive. However, there are some compelling reasons why THz is a better option. The advantages of THz wireless include aiming stability, responsiveness to atmospheric turbulence, and eye protection. They enhance each other's performance in inclement weather: In the rain or fog, THz performs better [3]. Tables 5.3 and 5.4 show different mid and high bands for wireless communications in 6G.

5.3.3 Optical Wireless Technology

OWC technologies are foreseen for 6G communications for any potential device-to-access connections; such networks also enable routing network access. Since 4G communication networks, OWC technologies have already been employed. However, to fulfill the needs of 6G communication networks, it is anticipated that they will have to be employed more broadly. Light fidelity, visible light communication (VLC), FSO communication optical, and camera communication based on the optical band are examples of OWC innovations that are well adapted. Pieces of research are focused on improving the performance of these technologies and solving their limitations. Wireless optical communications should deliver increased data speeds, low latencies, and secure connections. LiDAR is a revolutionary technology for very-high-resolution 3D mapping in 6G networking, as it is likewise based on the optical band. OWC is convinced that mMTC, uHSLLC, uMUB, and HDD technologies will be better supported in 6G communication networks. Advancements drive the OWC in 6G in light-emitting-diode (LED) technology and multiplexing methods. In 2026, both microLED technology and spatial multiplexing techniques are predicted to be robust and cost-effective.

Table 5.3 mmWave, THz, and optical sub-bands for possible wireless communications in 6G [7].

	mmWave part-1		30–275 GHz	10–1.1 mm	mmWave	Optical
THz	mmWave part-2		275–300 GHz	1.1–1 mm		
	Far IR part-1		0.3–3 THz	1–0.1 mm	Infrared	
	Far IR part-2		3–20 THz	0.1–0.015 mm		
	Thermal IR	Long-wavelength IR	20–37.5 THz	0.015–0.008 mm		
		Mid-wavelength IR	37–100 THz	0.008–0.003 mm		
	Short-wavelength IR		100–214.3 THz	3 000 000–1400 nm		
	Near IR		214.3–394.7 THz	1400–760 nm		
	Red		394.7–491.8 THz	760–610 nm	Visible light	
	Orange		491.8–507.6 THz	610–591 nm		
	Yellow		507.6–526.3 THz	591–570 nm		
	Green		526.3–600 THz	570–500 nm		

	Blue		600–666.7 THz	500–450 nm		
	Violet		666.7–833.3 THz	450–360 nm		
	UVA		750–952.4 THz	400–315 nm	Ultraviolet	
	UVB		952.4–1071 THz	315–280 nm		
	UVC		1.071–3 PHz	280–100 nm		
	NUV		0.750–1 PHz	400–300 nm		
	Middle UV		1–1.5 PHz	300–200 nm		
	Far UV		1.5–2.459 PHz	200–122 nm		
	Hydrogen Lyman-alpha		2.459–2.479 PHz	122–121 nm		
	Extreme UV		2.479–30 PHz	121–10 nm		
	Vacuum UV		1.5–30 PHz	200–10 nm		

Table 5.4 THz channel features and impacts on 6G communication systems [8].

Parameter	**Dependence on frequency**	**Impact on 6G THz systems**	**THz vs. Microwave and FSO**
Spreading loss	Quadratic increase with decreasing area and constant gains; quadratic decrease with constant area and frequency-dependent gains	Distance limitation	Higher than microwave, lower than FSO
Atmospheric loss	Frequency-dependent path loss peaks appear	Frequency-dependent spectral windows with varying bandwidth	No clear effect at microwave frequencies, oxygen molecules at millimeter wave, water and oxygen molecules at THz, water, and carbon dioxide molecules at FSO
Diffuse scattering and specular reflection	Diffuse scattering increases with frequency; specular reflection loss is frequency-dependent	Limited multi-path and high sparsity	Stronger than microwave, weaker than FSO

Parameter	Dependence on frequency	Impact on 6G THz systems	THz vs. Microwave and FSO
Diffraction, shadowing and LoS probability	Negligible diffraction. Shadowing and penetration losses increase with frequency. Frequency-independent LoS probability	Limited multi-path, high, sparsity and dense spatial reuse	Stronger than microwave, weaker than FSO
Weather influences	Frequency-dependent airborne particulates scattering	Potential constraint in THz outdoor communications with heavy rain attenuation	Stronger than microwave, weaker than FSO
Scintillation effects	Increase with frequency	Constraint in THz space communications	No clear effects at microwave; THz less susceptible than FSO

5.4 Extreme Performance Technologies in 6G Connectivity

Substantial bandwidth and extremely directional antennas will be available to 6G handheld devices via the millimeter-wave (mmWave) and terahertz (THz) frequency bands, enabling software innovations and uninterrupted connectivity. The use of incredibly high computing power processors at both the infrastructure and smart objects will substantially reduce latency in 6G [9]. The upcoming platform's cell devices will be intelligent enough to sense the environment and take cautious and preventative steps. To enable the key characteristics of 6G, a combination of cutting-edge technologies should be used [10].

5.4.1 Quantum Communication and Quantum ML

Quantum technology makes ultra-accurate clocks, diagnostic imaging, and neural networks possible by utilizing quantum mechanics' features, such as the interaction of atoms, molecules, or even particles and electrons. The maximum potential, however, has yet to be realized. A quantum interconnection that connects quantum computers, models, and sensors to distribute resources and information around the world securely. The first service to use this infrastructure will be quantum key distribution. It offers an inherently secure random key to both the server and client of an encrypted message, preventing an adversary from eavesdropping or controlling the system. It will protect crucial sensitive communications from future intelligent computers breaking codes. This should offer a safe exchange of information, digital certificates, digital signatures, and time synchronization, among other things [11]. This infrastructure would benefit the economy and society while also ensuring important government data from the ground, sea, and orbit. QC and QML may handle complicated challenges like a multi-object exhaustive study in ad hoc routing protocols and Cloud IoT by delivering rapid and best-prop selection to information in the communication network backbone.

5.4.2 Blockchain

Blockchain is revolutionizing some of the world's most important businesses, including finance, operations management, accounting, and transnational remittances. The blockchain concept is bringing up new ways to do business. Blockchain establishes trust, openness, confidentiality, and autonomy among all participants in the network. The most crucial characteristic for successful enterprises in the telecommunication industry is innovation in a competitive environment with lower costs. The telecommunications business can profit from blockchain in several ways.

5.4.2.1 Internal Network Operations

Ethereum, the second generation of blockchain technologies, has transformed the automation system in various applications with smart contracts. When a specific event occurs, smart contracts allow mathematical equations to execute automatically. Because of this, blockchain has a lot of interest in the telecommunications business, where it may be used to automate things like billing, systems integration, and roaming. Blockchain can prohibit malicious traffic from entering the network system, saving a lot of bandwidth and expenses and, as a result, increasing the carriers' revenue. The telecommunications industry can save time by minimizing the time-consuming post-billing audit procedure by using smart contracts for bill validation and clearance. The telecommunications industry can automate auditing and accounting operations using this method [12].

5.4.2.2 Ecosystem for Productive Collaboration

The goal of next-generation wireless systems is to offer a wide range of new digital services. For the complicated transactions required for these services, blockchain is an appealing application. By efficiently exploiting user information, blockchain can potentially be employed in the advertising sector. Vast M2M interactions will emerge as a result of this. Telecommunications companies may lead the way in this sector by implementing blockchain technology and ushering in the next wave of digital operations. The desire for a huge connection in 6G has prompted system resource management difficulties such as power distribution, spatial multiplexing, and the deployment of computation power. Using smart contracts to manage the connection between operators and users, blockchain can bring solutions to the 6G network in several fields. Consequently, blockchain can help with spectrum power and risk management issues that aren't covered by licenses. In addition to flawless environmental protection and surveillance, blockchain can be employed in e-health and decrease cybercrime rates.

5.4.2.3 Tactile Internet

With the advancement of internet technology, data and social networking has become possible on mobile devices. The evolution of the Internet of Things (IoT), in which interconnection between smart devices is facilitated, is the next phase. Tactile internet is the next stage on the internet of networks, which incorporated real-time M2M and H2M communication by adding a new element of haptic sensations and tactile to the mix. The term *tactile internet* refers to touch transmission over large distances [10].

5.4.2.4 Spectrum Sharing (FDSS) and Free Duplexing

Previous wireless generations utilized fixed duplexing (TDD/FDD) or flexible duplexing (FDD) in the instance of 1G, 2G, 3G, and 4G, and flexible duplexing in the context of 5G. With the advancement of duplexing technology, 6G is expected to adopt a full free duplex, where all users are permitted to use all resources simultaneously. As a result, as AI and blockchain are projected to be major technologies in 6G, strong spectrum monitoring and spectrum management procedures will be established for the 6G rollout. AI-assisted 6G systems can dynamically control network resources. As a result, in 6G, open spectrum sharing will become a possibility.

Even though 6G is only a couple of years away, few service providers are currently considering it. However, as we pinpoint where 5G misses, 6G exploration is projected to ramp up. It won't be long until the forces that begin debating how to go about it, as 6G will improve on the unavoidable flaws and shortcomings of 5G. 6G network will mirror every advance that 5G delivers and, even better, enhanced version. As we've experienced with 3G, 4G, and 5G in the past, as a channel's capacity grows, so do its applications. This will have a huge impact because new goods and services will fully leverage 6G's throughput and other advanced capabilities.

5.5 6G Communications Using Intelligent Platforms

Society is becoming increasingly digitalized and will continue to do so into the future. The world is moving toward the deployment of 5G wireless networks. Wireless connectivity is now ubiquitous to connect people, and things allowing machines to communicate without the need for human intervention. 5G New Radio (NR) network should be able to handle high connection density and high user plane latency. 3GPP Release 15 defines the basic framework for NR, which supports ultra-low latency transmission through a very short mini-slot. The next release of 5G will feature enhanced URLLC (ultra-low latency connectivity), mMTC (massive Machine Type Communications), and NOMA capabilities, and NB-IoT and LTE-M will be integrated into 5G NR for MTC services. LPWANs such as SigFox, LoRA, and Ingenu have been developed for MTC applications. The 5G network can be used to provide mMTC and URLLC services, though the data rates are rather limited and the supported use cases are rather limited. To ensure robustness, connectivity solutions for MTC scenarios may require multiple radio access technologies. Although 6G research is still in the exploration phase, many publications have already been published on what 6G will be.

MTC is expected to be a key driver in the increasing trend toward making everything that moves autonomous and intelligent. 6G is enabling super smart cities with 30 billion connected devices by 2030. Industry 5.0 will involve more interactivity and novel human–machine interactions than Industry 4.0 and will be supported by MTC. MTC will play a fundamental role in facilitating the user experience of augmented/virtual/mixed reality, and in enabling novel and efficient man–machine interfaces. MTC will need to be energy-efficient, low-latency, and error-free to enable zero-energy, zero-touch systems. In 6G, data marketplaces will link data suppliers and customers and will add new KPIs like age of information, privacy, and localization accuracy. Industry 4.0 will require factories to be flexible, agile, and adaptable, supporting both automation and human-machine interface. Swarms of autonomous vehicles will be commonplace in 2030, requiring robust connectivity solutions to perform in such an environment. Wearable devices will be seamlessly integrated into our clothing in the 6G era. At the edge of the IoT, machines are low-power sensors that can keep going forever.

Wireless energy transfer and backscatter technology could make this possible. Human senses can interact with machines by using ultra-broadband and ultra-responsive network connectivity. Decentralized ledger technologies (DLT) will be used to transfer data and micropayments between IoT devices. As 6G becomes more prevalent, 5G will require more stringent requirements. In 6G, the ultimate goal is zero-energy MTDs, achieved through a combination of efficient hardware design and energy harvesting techniques. New KPIs are becoming increasingly relevant for 6G and relate to the Age of Information (AoI), interoperability, dependability, positioning, sustainability, and E2E EE. Seamless integration of heterogeneous network access technologies will help 6G technologies to provide reliable service for emerging applications. The EE vision for 6G will be wider and more stringent than the EE vision for 5G. In 5G, URLLC (critical) and mMTC (large/dense MTC) were established for small payloads, low-data rates, and sporadic traffic patterns.

In the coming decade, 6G will have to be scalable and multidimensional. To successfully meet 6G visions, novel approaches are needed to design enabling technology components. The MTC network architecture is meant to harmonize connectivity across the network and maximize performance. The MTC network architecture needs to accommodate a number of key trends affecting the MTC network architecture that including the evolution from cell-based networks to cell-free networks, the appearance of local micro-operators deploying new private or public networks, and the increasing number of traffic patterns and service classes.

The network architecture of the future should allow for the orchestration of applications across multiple heterogeneous networks and the use of technology-agnostic interfaces. The selection of the RAT(s) and the configuration parameters of the RAT(s) should be handled using machine-learning methods. Machine clients can change their traffic patterns in different places, and this postulates the need for centralized or distributed optimization mechanisms. Introduce manageable control and emergency traffic channels within the MTC architecture and ensure backward compatibility of new generations and releases within one generation. Networks require a new air interface to power zero-energy devices. The development of the underlying optimization mechanisms and technical solutions for the post-5G MTC network requires the development of common interfaces and procedures. Ultra-low-power (ULP) transceivers are essential for mMTC to operate in a sustainable way. Significant reductions in energy consumption can be achieved by considering the energy consumption of the MTD as a whole. The power consumption of the various building blocks of the transceiver can be reduced by better power schemes of the various building blocks.

5.5.1 Integrated Intelligence

Unlike previous generations of wireless communication technologies, 6G will be revolutionary, moving wireless technology from “devices connected” to “connected competence.” Any step of communication, along with radio resource planning, will be automated with AI. AI's widespread implementation will usher in a new era of communication networks [11]. For ultra-dense complicated network scenarios of 6G, a complete AI system is required compared to 5G, empowering intelligent electronic devices to obtain and undertake the resource distribution operation [13].

5.5.2 Satellite-Based Integrated Network

To enable ubiquitous connection, satellite communication is required. It is almost unrestricted by geographical conditions. It may offer a seamless worldwide coverage of various geographic places such as land, sea, air, and sky to serve the user's seamless computing. To reach the goal of 6G, it is expected to link terrestrial and satellite technologies to deliver always-on broadband worldwide mobile connectivity. 6G will require integrating terrestrial, satellite, and aerial networks into a unified wireless network [13].

Each new generation of technology has its advantages, alongside its challenges that must be overcome before the technology can be fully realized. With technologies up to 4G focusing mostly on capacity and coverage, 5G wishes to push those boundaries, increasing connectivity and capacity while simultaneously further reducing power consumption as well as latency. 5G offers many advantages, but its challenges and restrictions are incredibly stringent and require existing infrastructure to be pushed to, or beyond, its limits.

4G functions on a 2D plane, where everything is flat and easy to work with. 5G and by extension 6G wish to function on a 3D plane, covering much more than the previous 4G-LTE generation was capable of covering. For this to become a reality, new frameworks and technologies will have to be created and put into place, starting with this so-called 3D architecture.

The 2D platform can be explained very simply as the networks that already exist on the ground here on Earth. This is everything from wireless access points in cities, cell and radio towers, and so on that provide internet and mobile coverage to the 2D plane of people on the ground. To introduce a 3D plane for internet connectivity, several technologies will need to be implemented to achieve the goal of lower power consumption, low latency, and high speed. These technologies include UAVs, LEO/GEO satellites, and more to bridge the gap between land, air, and space.

This new architecture is theorized to utilize a technology known as network slicing that is already being implemented for 5G networks. This technology works by splitting up existing networks and allowing for more customizability and use outside of just providing connectivity to those on this network. However, this slicing will be used not just across terrestrial nodes but over space and flying nodes as well.

Network systems have always had a standardization problem. For example, while most of the world utilizes their 4G networks and are standardized in that sense, the United States uses LTE technology instead, which is different in several ways from 4G and would potentially cause interoperability issues if the two were connected. For this new generation of technology to come to fruition, and its advantages be afforded to all, standardization across the board will have to occur. Without standardization, 6G and all tech connected to it will fail.

For the change from 5G to 6G to succeed, the 3D hierarchy and architecture must be implemented from the beginning. Enhancing existing systems and bolstering their capabilities while simultaneously creating new platforms of technology will be essential. The first obstacle that must be overcome is the ever-changing environment of connected devices. Stationary infrastructure will be incredibly inefficient at achieving the goal of 6G, so UAVs and satellites that can move to change their service area will be necessary. Of course, this has its challenges, too, such as attaining sufficient energy and overcoming interference from the atmosphere and other sources.

These UAVs will also be used to bring cloud services into the flying layer, as well as the space layer through the use of LEO/GEO satellites. By providing a UAV or satellite with sufficient computing power, field research and on-demand computations of datasets can be made easy by sending data to these UAVs, allowing built-in cloud capabilities to do the necessary processing and then retrieve the results, allowing for mobile research bases and other mobile data-collecting ventures to be much easier and more fruitful. This also brings along with it the idea of AI being distributed over this same sort of network. Having access to a pervasive and on-demand AI service could be beneficial for similar use cases.

The biggest obstacle alongside technological limits as well as funding for such a venture to create this new 3D network hierarchy will be interference. Satellites and other low-orbiting signal-producing apparatus have to contend with interference on a constant basis already, and with a more pervasive web of such devices being the future of 6G, that interference will only increase. To combat this, it is proposed to exploit the classification system of interference, as well as other technologies to detect interference before it happens to change transmission power at precise times to avoid interference such as this.

The internet of things, artificial intelligence, generations of various speeds, and satellites all began in their own little corner of the technology spectrum. Each advancing over time but when they started working together, coexisting through intelligent connected environments they became something greater than themselves. These combinations have revolutionized the technology industry and additional industries where technology is used – which is basically everywhere, by the way. Compacting these types of advancements into small satellites alone is leading to the creation of artificial constellations all managed and operated through the technologies through everything listed above. Satellites are relied on across the world for nations to operate successfully, generating information, regulating trade, and looking toward the stars' advanced satellites could potentially be the pinnacle of technology altogether.

5.5.3 Wireless Information and Energy Transfer Are Seamlessly Integrated

6G wireless networks will also send power to recharge battery devices such as smartphones and sensors. WIET (wireless information and energy transfer) will be incorporated as a result. Universal super 3D networking: Drones and very low Earth orbit satellites will access the network and key network functions, making super-3D compatibility in 6G ubiquitous.

5.6 Artificial Intelligence and a Data-Driven Approach to Networks

With 6G expected to bring humanity further toward autonomous applications, the question of protecting personal information (PI) comes into play. How is the data used, and who uses it? The challenge is quantifying what identified data means to develop proper measures. It becomes a tradeoff between managing privacy and building the required trust. The more trust that is given the more risks of privacy leaks grow. This means 6G will require new trust models to balance between maintaining customers' trust and privacy.

PI and personally identifiable information (PII) are used interchangeably in legislation frameworks. The date of birth is considered PI but not PII as it can't identify a single person. The question that needs to be answered is, “How many features must be linked before PI becomes PII for an individual known to be in a dataset?” With the definition of an individual's identity being very broad, it in principle covers any information that relates to an identifiable individual during their lifetime or decades after their death. As the more PI data is linked, the larger the personal information factor (PIF) becomes. Data above a specific threshold means that sufficient PI exists to identify an individual. Since there is no way to unambiguously determine when linked when the dataset crosses the threshold, it has become a major problem for many different technologies like AI and IoT. This has significant consequences for all smart services as a whole. This is still an issue unaddressed in 5G, and it is impossible to be ignored for the future of 6G

As 5G evolves, the reliance on AI smart applications increases. Due to the number of challenges for the future of wireless applications is the implementation of distributed ledger technology (DLT). DLT technologies such as blockchain provided security and privacy features such as immutability and anonymity. Privacy protection that uses differential privacy seems promising when addressing challenges that could arise in the future 6G wireless applications as it operates by perturbing the actual data using artificial designs.

Protection of individual privacy has been a challenge for standardization since nations have different perspectives in their area. International committees such as the joint technical committee 1 (JTC 1) are working on different privacy frameworks, but much more needs to be done. Due to the interconnectedness of the global markets, “Privacy by Design” will most likely prevail in newer products and communication techniques.

A future that sees us utilizing digital avatars will also have health and safety concerns. The future digital and physical worlds will become entangled with malicious cyberattacks, possibly leading to loss of property and life.

With 6G in 2030, AI has been proposed to become fully integrated with the communications backend, handling many time-consuming tasks like network selection, handoff or handover, and resource allocation. This will allow the network to dynamically adapt to changing usage conditions while tailoring an experience for each individual user. The improved specs and capabilities of 6G are also set out to change the industry in our world. The higher data rates and user densities possible prove promising for all sorts of technologies like extended reality (a new facet of augmented reality), production automation through robotics and other autonomous systems, wireless brain integrated computer interfaces, and high-performance real-time healthcare for anyone in the country. The new workings of 6G will also promote wireless power transfer. Think wireless charging, but without a charging pad. Now, your phone could very well charge in your pocket as you walk down the sidewalk.

Table 5.5 Artificial intelligence data rates and latency features [13].

Peak data rates	100 Gbps to 1 Tbps
Extreme ultra-low latency	0.1 ms
Density	100 devices per m^3
Extreme ultra reliability	Max. 1 out of million outage
Extreme coverage	Sea, sky, space

Artificial intelligence technology is growing from simple tools used by average scientists to as far as to use within the professional development community for higher intelligence use. These various organizations can use AI technology to fill the current gap in the data science area, which could be a big game-changer for data science! This also always gives end-consumers the ability to take their business and personal data with them wherever they go. This, of course, is a big deal when it comes to making end-consumers happy while retaining their privacy with their online data.

When it comes to data processing with the help of 5G and 6G, AI, and IoT devices, modern technology and its data often demand that data management and processing capabilities have the data possessors brought to the data itself rather than sending the collected data off for processing elsewhere. This may also include the data processing and distributed data stores all to be included within the data management process to guarantee support is offered where the received critical data is being stored. 5G network technology will be revolutionary to end consumers, businesses, as well as data processing centers around the globe. Table 5.5 shows some features of AI.

5.6.1 Zero-Touch Network

Trust extends across all protocol layers, from IP to apps and data. In order to answer issues like: can this host interact with a remote party without being attacked or hacked, a trust system in network communication is needed. To sense, comprehend, and program the world, 6G will create a wide digital/physical world boundary. As a result, a breach of information security can put people's physical safety at risk and cause property loss, in addition to causing data loss, control loss, and financial loss. International wars could deploy foreign cyberwar warriors to wreak such damage in a country that traditional warfare is no longer necessary to persuade the victim to accept the attacker's demands. You can use trust networking to give remote access to specialized 6G networks or any other important infrastructure in local or national packet data networks (PDNs). The PDN service could be made available to customers. Last but not least, trust networking can be utilized across all platforms. Examples of use cases range from a simple, single administration handling the entire trusted network and all of its devices to cases where devices are owned by independent parties, and different layers, segments, or subsystems of the network are owned and managed by numerous different stakeholders with potential conflicts of interest. A mobile or IP network's adoption of trust networking calls for alterations to the existing network. Those changes should be able to be implemented one by one in a network. If device and network changes are required, they can really only be made by a single administrator. When a server can certify client software and vice versa, it's a good example of a single admin solution.

Networks are becoming increasingly complex as new technologies such as 5G, smart devices, and Cloud finds legitimacy. Humans are finding it difficult to keep up with increasing complexity and maintain networks running at the level required by evolving services. We have already been inundated with data, and it is only the beginning. Running a 5G network, incorporating data points from the IoT, while facing the challenge of mission-critical use situations would be only conceivable whenever AI, robotics, and Big Data are used to drive telecom networks' “Big Data operations” [14].

We are making progress toward our dream of “zero-touch networks” thanks to data-driven processes. The zero-touch network is an application network service that uses automation and AI to supplement the human experience in its operations. To realize the idea of zero-touch connectivity, systems and capabilities must operate together. Because a zero-touch network is built on intelligence, it necessitates the implementation of network elements and operational services that support AI technology. The system's products and services must be created from the ground up with intelligence in mind. Automation is a critical component of the overall strategy. Intelligent agents are software that can work alone and make intelligent decisions in a complex environment [10].

5.6.2 AI by Design

According to the technology strategy, AI will be used across the service or product portfolio in all components of the system architecture. The goal is to enable humans and automated systems to evolve an engineered and responsive system into a lifelong learning network that better meets the needs of our customers.AI can be used to solve specific challenges for telecommunication companies, creating value where everything matters most. It is not a broad-sense tool designed for every conceivable use case [15]. AI is being developed to address today and tomorrow's mobile communications challenges. Increasing the return on installed base without the need for site visits or the addition of new equipment. Providing consumers with always-on, high-performance networks. Providing mission-critical corporate infrastructure for national services. Adhering to elevated customer expectations and providing an impeccable QoS journey for enterprises. Acknowledge the importance of the Cloud by improving agility, lowering costs, and laying the groundwork for additional technologies. Managing growing demands while preserving operational expenses down [10]. Creating flexible infrastructures based on real usage and guaranteeing that it is kept to a minimum.

5.6.3 Technological Fundamentals for Zero-Touch Systems

6G components in communication systems need to embrace artificial intelligence from the onset to capitalize on the power that machine learning can infuse into technological advancements. An automation and strategic planning solution includes automated foresight operations, neurological risk mitigation, and data analytics for confidence, conscience networks, and innovative caseload positioning. The AI-based switchover, 5G-aware congestion management, evolutionary computation offloading at launch, uplink-traffic stimulated movement, transport network control, and amplified MIMO sleep are all features of cellular networks [16]. Deep learning paging on the cloud core is a premium feature. Automation is a critical component of the workaround. An intelligent agent is an application that can operate autonomously and make smart decisions in a dynamic environment.

Forward-thinking telecom companies and virtual network operators need an asynchronous network and commercial enterprise platform to allow them to switch to a new business reality characterized by 5G, the IoT, network virtualization functions, and application networks. Traditional ML systems offer numerous advantages to telecom service providers, especially in maintaining effective levels of quality. Notwithstanding, in addition to leaving a large network presence, the huge data transfer associated with traditional ML models can be problematic in the context of data protection [16]. Federated learning (FL) addresses these issues by bringing “each encoding to the data” rather than transmitting “the data to the encoding.”

The idea of “intellectual purpose management,” which uses innovative AI, is growing in popularity. Service providers and their end users will clarify “what” actions individuals need from the system. The framework will then determine “how” to achieve this query by employing closed-loop digitization and as little user intercession as feasible. We are investing heavily in robotic systems and enterprises to facilitate these product features in the value propositions.

5.7 Sensing for 6G

5.7.1 A Bandwidth as Well as Carrier Frequency Rise

The 6G would continue the trend started by 5G. NR networks continue to push for higher frequency ranges, bigger bandwidths, and larger antenna arrays. As a result, sensing solutions with the precise spectrum, harmonic and directional resolutions, and localization to cm-level precision will be possible. New materials, device kinds, and other innovations are also on the horizon [17].

Network operators will be able to meld and manipulate the electromagnetic field via customizable surfaces. Simultaneously, ML and AI will become more prevalent to take advantage of the extraordinary availability of data and computational resources to tackle the most difficult and complex problems. High carrier frequency RF-based sensing will enable more accurate approaches to measuring the surroundings, and detecting and recognizing things. It provides a larger spectral range chance to monitor and track new types of objectives and parameters that aren't detectable in conventional methods of radio spectrum currently in use [17].

Small cell deployments will begin to dominate as cellular systems evolve to mmWave bands in 5G and potentially sub-THz bands in 6G. Wide bandwidth systems deployed in cellular network architectures provide a practically unseen potential to use the data connection for sensing. 6G systems will indeed be highly smart wireless platforms that will enable extremely accurate localization and leading provider services in addition to widespread communication. They will be the driving force behind this popular revolt by introducing a completely new set of features and service capabilities. Localization and sensing would also intermingle with interactions. It is constantly able to share readily accessible resources in time, rate, and space [18].

We can identify four emerging technological enablers for 6G communication networks and localization and sensor technology:

1. The use of innovative wireless communications frequencies
2. Incorporation of an intelligent substrate surface
3. Smart beam-space handling, AI
4. Deep learning techniques

The transition to THz frequencies has several significant advantages in terms of localization and sensing. For starters, stimuli at these frequencies cannot permeate objects, resulting in a much more direct relationship between both the transmit power and the propagation environment. Furthermore, greater absolute data rates are obtainable at high frequency, resulting in far more correctable multi-path in the latency domain with more highly reflective components.

Finally, shorter wavelengths presuppose smaller antennas, allowing discrete components to be stocked with dozens or hundreds of transmitters, which is advantageous during angle estimation. This is useful across both passive and active sensing. To obtain the desired output, chip technology that supports cost advantages must be available. Furthermore, suitable channel designs that effectively classify the proliferation of 6G waves out over firmware and the air will be needed to facilitate innovative alternatives and methodologies.

5.7.2 Chip Technologies of the Future

There is now a concise need for further technological development to support the frequency mentioned above bands cost-effectively. One critical aspect is incorporating the necessary technology [19]. Presently, radio systems designed to operate throughout the numerous different 100 GHz normally entail antennas and signal processing facilities, which are outrageously massive to embed into conventional sensor nodes. While the front-end and wireless connection chips have taken time to develop, the processing method of preference is now sufficient to allow effective propagation, down-conversion, and further signal handling, even at huge frequency bands.

5.7.3 Models of Consistent Channels

Models of electromagnetic wave propagation are essential for the proper design, function, and efficiency of cellular networks. Indeed, when designing new heuristics or deploying new architectural design, modeling available radio complexities and process controls becomes critical for analyzing the overall effectiveness of mobile telecommunications frameworks. Even though it is difficult to build sonar, sensing, ranging, or direction approximation algorithms without audio suppositions on how microwaves perpetuate in the proximity of the sensor system, propagation models are equivalently vital to ensure reliable and efficient aligning. Frameworks are also effective in predicting and benchmarking the correctness of multiple sensing and localization techniques. If measured by standardized radio signals, greater systems are far more vulnerable to atmospheric conditions and do not continue to spread properly via the materials. These characteristics are also important for sensing.

Although we reach higher wavelengths and spectroscopic assessment, we should best understand and template the permeability of electromagnetic fields via various molecules and the reflection properties of various substances. Identifying the perfect band for the right frequency to maximize any use of prospective future applications is a major challenge. As we move closer to the data-rich 6G era, expert systems will become highly significant. A wide-ranging field of study establishes artificial intelligence and agents capable of achieving rational aims and objectives based on reasonable and predictive reasoning, prepping, and effective strategic thinking, possibly in dynamic situations. AI systems are typically built on machine learning, which further offers data-driven collaborative learning approaches frameworks that go beyond formal presuppositions [20].

5.7.4 X-Haul and Transport Network for 6G

Networks have difficulty providing extremely high data speeds (tens of gigabits per second) with very low latency (a few milliseconds). These demanding criteria occur when telecom operators' average income per user is declining, emphasizing the need for alternatives that can lower their operational costs to a competitive threshold. Shaul, dubbed “the option open transport method for future 6G networks,” seeks to combine fronthaul and backhaul infrastructures, as well as many of their fixed and wireless technologies, into a single packet-based transportation system under SDN- and NFV-based (network functions virtualization) common control [21].

Over ubiquitous 3D coverage areas, 6G mobile technology is expected to enable data throughput of 100 Gbps or greater, as well as ultra-low latency. The transport network, which primarily links the cell sites to the network core, has not kept up with the cell sites. Consequently, it has been identified as a potential barrier to the implementation of high-performance and cost-effective networks. As a result, to meet the new and varied requirements of

6G and even beyond mobile networks, which are being driven by the rollout of large numbers of mobile cell sites, extremely diversified architectures, and complex services and applications [21].

The vehicular infrastructure has been built as an effective instrument to link human transportation worldwide for many years to come. However, as the number of vehicles increases, the vehicular network becomes more homogenous, vibrant, and massive. This will make it much more difficult to address the rigorous standards of the next-generation (6G) network, including ultra-low latency, higher performance, heavy stability, and enormous interconnection.

Lately, ML has evolved into a powerful artificial intelligence methodology for increasing the efficiency and adaptability of either machine or wireless technology. Inevitably, incorporating ML into spectrum sharing and network is a contentious issue that is being extensively researched through industry and academia, clearing the way for the new intelligent action throughout 6G sensor communications.

Every new generation of communication technology introduces new and intriguing features. The 5G communication technology, which will be formally implemented around the world in 2020, has many unique characteristics. However, in 2030, 5G will not fully serve the expanding demand for wireless communication. As a result, 6G must be implemented. 6G research is still in its early stages. With both the readiness and impending commercial exploitation of 5G network technologies, an abrupt change in cellular data consumption, as well as the variable requires of vertical markets such as innovation, e-commerce, public transit, and health, can sometimes be assisted as part of a much larger global digitalization revolt. In the meantime, the installed capacity foundation of the IoT network devices, such as computers and phones, is expected to reach 75.44 billion through 2025, representing a fivefold spike in a couple of years. As wireless networks reach their physical size restrictions, this transition will pose a massive challenge to future wireless communications. Concerns are also growing about the considerable demand for energy to maintain such systems [21].

As a result, it is critical to creating new innovative platforms to improve this growing number of devices in smart, energy-efficient forms. To recognize this, modern world 5G and beyond (6G) mobile systems will be a game-changer for socioeconomic development.

5.8 Applications

The characterization of 5G revolves around latency, throughput, reliability, energy, and costs. Various configurations of 5G networks address mobile broadband and latency issues. However, the configurations of 6G aim toward meeting stringent network demands in a holistic sense. Hence, their architectures are quite different from the 5G networks because they chase increased reliability and efficiency and reduce latency. Thus, 6G offers particular uses based on the network's characteristics. Figure 5.3 shows some future use cases for 6G.

- Augmented reality (AR) and virtual reality (VR) Since the inception of 4G networks, streaming and multimedia services have increased in the spectrum and needs because of the ever-developing data-hungry applications. Consequently, 5G developed new systems such as mmWaves to sustain the higher capacities. However, the “rooms for improvement” for the 5G network is presented when applications requiring multi-Gbps are developed. Hence, 5G essentially paves the way for the early adoption of augmented reality and virtual reality. Full adoption of these two technologies will overpower the 5G spectrum because they require a capacity of approximately 1 Tbps. This capacity can only be achieved by developing 6G networks because 5G has a limit of 20 Gbps. Moreover, the latency requirements for 5G cannot effectively compress augmented reality and virtual reality.

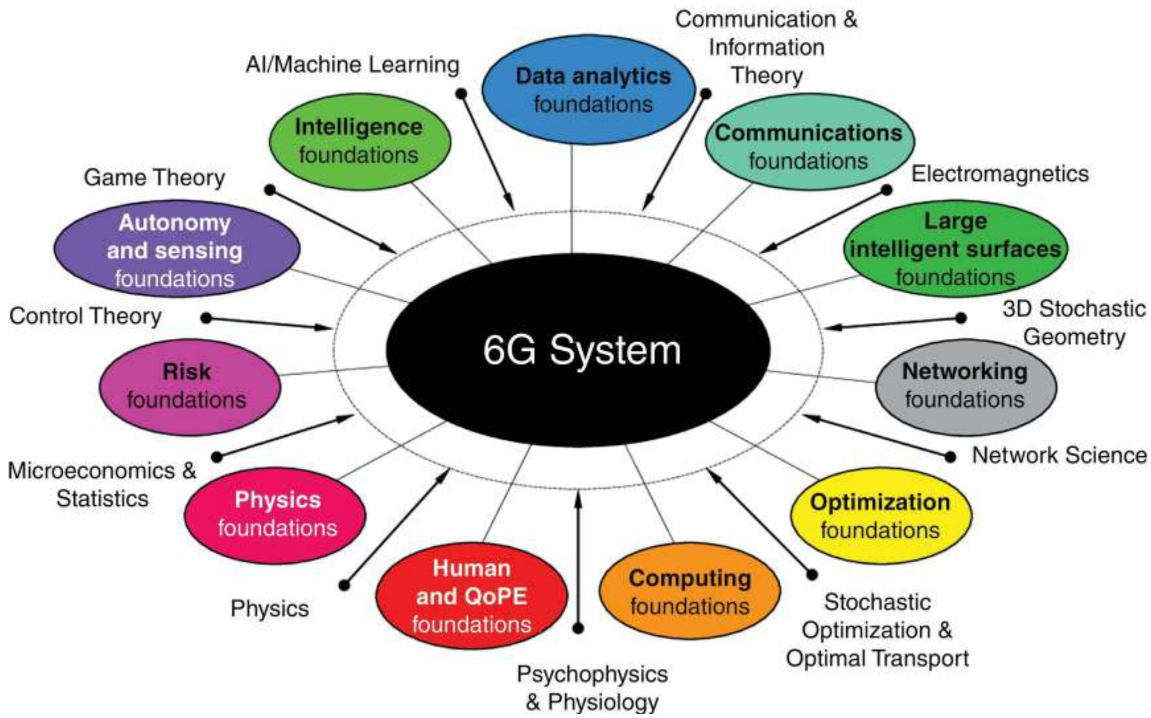

Figure 5.3 The use case for 6G technology [22].

➢ eHealth

The healthcare sector will greatly benefit from 6G because of telehealth and telemedicine technologies. These technologies are critical in a fast-evolving world because doctors can dispense healthcare services such as surgery and pharmacy remotely to their patients. Such eHealth services demand QoS. The QoS requirements includes stable and continuous connectivity, mobility support, reliability, and ultra-low latency. Thus, developing 6G networks that can achieve such requirements enhances the full actualization of eHealth services. This is because the increased spectrum and the combined intelligence of 6G networks will ensure spectral efficiency.

➢ Pervasive connectivity The degree of growth of mobile technology is very high, meaning that mobile traffic is on the rise. Research estimates that approximately 125 billion devices will be connected by 2030. Based on such high estimates, it is probable that 6G network transfigurations will support such extremes. For instance, 6G will connect all smart and personal devices. Consequently, this makes them require high-energy frequencies to enhance scalability and better coverage.

- ➢ Industry 4.0 and Robotics Technology specialists project that 6G is likely to materialize the 4.0 industry revolution initiated by 5G transfigurations. Industry 4.0 is characterized by physical (cyber) systems and the services of the internet of things. 6G will be cost-effective for industries because their architectures are built to increase the cyber computational space. Furthermore, 6G materializes reliable isochronous communication, which is critical in addressing upcoming disruptive technologies.
- ➢ Unmanned mobility Connecting autonomous vehicles and machines will be one of 6G's major focuses. This means that data rates will increase, thereby demanding high levels of reliability and low latency. For instance, it is estimated that reliability should be about 99.99999%, whereas latency rates should be below 1 ms. Besides, safety is also a key factor when dealing with unmanned mobility because of the increasing data rates. Drones currently have practical uses in the military and science, but their technology is being explored for use in several industries. These drones are often unmanned, and they are controlled and connect various systems of the world.

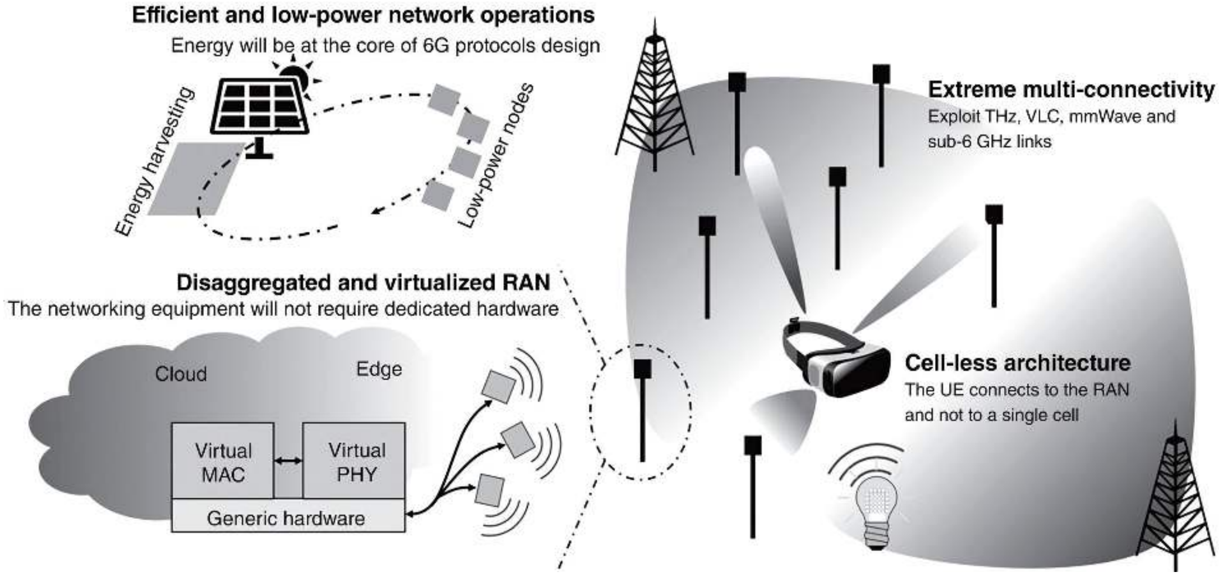

Figure 5.4 Architectural innovations of 6G networks [5].

5.9 Innovative 6G Network Architectures

6G will be characterized in many ways as substantively advanced as compared to 5G in prior versions of wireless evolution including: Terahertz-level data, extreme focus on short-range communication, AI is inherent within the network, extremely high radio density, very high network heterogeneity (extreme HetNets), and transformed radio topology (mesh, hops, and peer-to-peer). These characteristics will be manifest in the technologies, capabilities, and ultimately the applications, services, and solutions that distinguish 6G from all generations that preceded it. Figure 5.4 shows the architectural innovations of 6G networks.

The following paradigms are likely to be introduced and deployed in 6G architectures:

- Tight inclusion and integration of various frequencies and communication technologies and cell-less architectures
- 3D network architecture
- Disaggregation and virtualization of network equipment
- Advanced access-backhaul integration
- Energy-harvesting approaches

5.10 Conclusion

Disruptive communication technologies are coming. As suggested in this chapter, innovation and technology are inevitable. Thus, moving from 5G to 6G ensures that extremely higher frequencies are achieved, such as the mmWave band to support the higher spectrum technologies. It means that 6G networks will be able to work effectively to limit or prevent disruptive technologies.

New architectural paradigms are necessary for the heterogeneity of future networks, making 6G an important element in telecommunications. 6G would push us into using terahertz. This means we could potentially get a band of 275 GHz to 10THz, which drastically compares to the 5G band of 5.15–5.85 GHz. 6G is also expected to have unlimited bandwidth, which would mean that there would be constant full speed. 6G would have microsecond latency, one terabit per second at a latency of around 100 μs –

about 50 times faster than 5G is. The architecture created specifically for 6G would allow all of this to be possible and all the promises of 5G would be met with 6G. The architecture of 6G would allow AI, specifically ML, to grow and mature because of the absolute power that 6G would have. Integration of intelligence in network systems, AI, and IoT is critical in advanced communication systems.

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